

Practical Problems (OS)

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1. Practical5b.c

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code:

```
GNU nano 8.7.1 practical5.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if (fork() == 0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    if (fork() == 0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

Output:

```
gayathri@Gayathri: ~  
gayathri@Gayathri:~$ nano practical5.c  
gayathri@Gayathri:~$ gcc practical5.c -o practical5  
gayathri@Gayathri:~$ ./practical5  
-rw-r--r-- 1 gayathri gayathri 327 Aug 12 07:02 Childwait.c  
-rw-r--r-- 1 gayathri gayathri 534 Aug 6 04:02 Copy1to0ther.c  
-rw-r--r-- 1 gayathri gayathri 94 Aug 7 15:22 Hello.c  
-rw-r--r-- 1 gayathri gayathri 562 Aug 6 04:00 Linux.c  
-rw-r--r-- 1 gayathri gayathri 223 Aug 12 08:08 MultipleChild.c  
-rw-r--r-- 1 gayathri gayathri 174 Aug 1 05:49 ParentChild.c  
-rwxr-xr-x 1 gayathri gayathri 16040 Aug 1 06:05 Pcpid  
-rw-r--r-- 1 gayathri gayathri 221 Aug 1 06:06 Pcpid.c  
-rwxr-xr-x 1 gayathri gayathri 16032 Aug 1 06:07 Pcrun  
-rw-r--r-- 1 gayathri gayathri 211 Aug 1 06:07 Pcrun.c  
-rw-r--r-- 1 gayathri gayathri 311 Sep 2 05:38 Signal.c  
-rwxr-xr-x 1 gayathri gayathri 16080 Aug 12 06:50 childfork  
-rw-r--r-- 1 gayathri gayathri 281 Aug 12 06:51 childfork.c  
-rw-r--r-- 1 gayathri gayathri 8164 Aug 26 05:43 command_history.c  
-rwxr-xr-x 1 gayathri gayathri 16176 Aug 6 04:03 copyfile  
-rw-r--r-- 1 gayathri gayathri 534 Aug 6 04:06 copyfile.c  
-rwxr-xr-x 1 gayathri gayathri 15992 Aug 13 04:09 exclp  
-rw-r--r-- 1 gayathri gayathri 179 Aug 13 04:10 exclp.c  
-rwxr-xr-x 1 gayathri gayathri 16096 Aug 27 13:47 fcfs  
-rw-r--r-- 1 gayathri gayathri 516 Aug 27 15:57 fcfs.c  
-rw-r--r-- 1 gayathri gayathri 205 Aug 27 08:30 fifo_Write.c  
-rw-r--r-- 1 gayathri gayathri 240 Sep 2 05:27 fifo_read.c  
-rw-r--r-- 1 gayathri gayathri 220 Sep 2 05:30 fifo_write.c  
-rw-r--r-- 1 gayathri gayathri 258 Aug 27 04:13 fifo_writel.c  
-rw-r--r-- 1 gayathri gayathri 90 Aug 7 15:36 fork.c  
-rw-r--r-- 1 gayathri gayathri 90 Aug 7 15:37 fork.c'  
drwxr-xr-x 3 gayathri gayathri 4096 Aug 27 05:11 github-c-files  
-rw-r--r-- 1 gayathri gayathri 0 Jul 29 07:03 hello.c  
-rw-r--r-- 1 gayathri gayathri 425 Aug 5 08:29 linux.c  
-rw-r--r-- 1 gayathri gayathri 314 Aug 13 04:43 orphan.c  
-rwxr-xr-x 1 gayathri gayathri 16080 Aug 13 04:12 pcec  
-rw-r--r-- 1 gayathri gayathri 193 Aug 13 04:15 pcec.c  
-rw-r--r-- 1 gayathri gayathri 543 Sep 2 05:07 pipe.c  
-rwxr-xr-x 1 gayathri gayathri 16208 Aug 13 05:12 practical3  
-rw-r--r-- 1 gayathri gayathri 403 Aug 13 05:12 practical3.c  
-rwxr-xr-x 1 gayathri gayathri 16128 Aug 13 05:15 practical4  
-rw-r--r-- 1 gayathri gayathri 357 Aug 13 05:14 practical4.c  
-rwxr-xr-x 1 gayathri gayathri 16216 Sep 2 07:59 practical5  
-rw-r--r-- 1 gayathri gayathri 503 Sep 2 07:58 practical5.c  
-rw-r--r-- 1 gayathri gayathri 403 Aug 13 05:10 practive3.c  
-rw-r--r-- 1 gayathri gayathri 1119 Aug 27 16:02 priority.c  
drwxr-xr-x 4 gayathri gayathri 4096 Aug 13 05:29 process-lab  
-rwxr-xr-x 1 gayathri gayathri 16000 Aug 12 06:41 processid  
-rw-r--r-- 1 gayathri gayathri 122 Aug 12 06:42 processid.c  
-rw-r--r-- 1 gayathri gayathri 664 Aug 18 14:45 program1.c  
-rw-r--r-- 1 gayathri gayathri 1733 Aug 18 14:51 program2.c  
-rw-r--r-- 1 gayathri gayathri 1213 Aug 27 16:06 round_robin.c  
-rw-r--r-- 1 gayathri gayathri 2157 Aug 27 05:48 s3.c  
-rw-r--r-- 1 gayathri gayathri 678 Aug 27 05:57 s4.c  
-rwxr-xr-x 1 gayathri gayathri 16160 Sep 2 05:45 sigchild  
-rw-r--r-- 1 gayathri gayathri 528 Sep 2 05:45 sigchild.c  
-rw-r--r-- 1 gayathri gayathri 551 Sep 2 05:48 sigpipe.c  
-rw-r--r-- 1 gayathri gayathri 419 Sep 2 05:50 sigseg.c  
-rw-r--r-- 1 gayathri gayathri 340 Sep 2 05:44 sigterm.c  
-rw-r--r-- 1 gayathri gayathri 848 Aug 27 15:59 sjf_nonpreemptive.c  
-rwxr-xr-x 1 gayathri gayathri 16040 Sep 2 05:30 write.c  
-rw-r--r-- 1 gayathri gayathri 254 Aug 13 04:20 zombie.c
```

Code:

Output:

Ex; kill -SIGUSR1 PID
Ex: kill -SIGTERM PID